

Automated Capillary Segmentation and Quantification in Nailfold Capillaroscopy Images: A Comparative Study of Traditional Image Processing and YOLOv8 Deep Learning

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Abstract—Nailfold capillaroscopy is a non-invasive clinical technique used to detect microvascular abnormalities that signal autoimmune diseases such as systemic sclerosis. Manual counting of capillary loops is tedious, subjective, and hard to standardise. This paper presents two automated pipelines for capillary segmentation and counting and compares them head-to-head on a 30-image dataset spanning normal (N1, N2) and pathological (S2, S3) morphological classes.

The first pipeline is built on classical image processing: region-of-interest (ROI) extraction, Contrast Limited Adaptive Histogram Equalisation (CLAHE), Hessian-based vessel enhancement, adaptive thresholding, morphological cleaning, and connected-component counting. Evaluated on ten normal N1 images, it achieves a Mean Absolute Error (MAE) of 1.90 capillaries and an RMSE of 2.74.

The second pipeline fine-tunes a YOLOv8 instance-segmentation model on manually annotated data. Evaluated across all 30 images, it reaches an overall MAE of 3.00, but excels on pathological images (S2 MAE=1.14; S3 MAE=1.90) while struggling on high-density normal images (N2 MAE=7.00). The results reveal complementary strengths and a shared bottleneck: annotation quality. Expert re-annotation is identified as the single most impactful next step.

Index Terms—nailfold capillaroscopy, capillary segmentation, capillary counting, CLAHE, Hessian vessel filter, YOLOv8, instance segmentation, Roboflow, systemic sclerosis

I. INTRODUCTION

The capillaries visible at the base of a fingernail—called the *nailfold*—form a regular pattern of hairpin-shaped loops in healthy people. A doctor can examine these loops under a low-power microscope to detect signs of disease. This technique, known as **nailfold capillaroscopy**, is one of the most sensitive early indicators of systemic sclerosis (scleroderma) and other connective-tissue diseases [1]. Key signs that a clinician looks for include fewer loops than normal (rarefaction), abnormally enlarged loops (“giant” capillaries), small haemorrhages, and disorganised or absent vascular regions.

Despite its clinical value, the analysis is almost always done by eye. The clinician counts the number of visible loops, notes their shape, and decides whether the pattern is normal or abnormal. This is slow (a single image can take several

minutes), depends on the examiner’s training and experience, and produces counts that vary between observers. Automated image analysis can make the process faster, more consistent, and available to centres that lack a capillaroscopy specialist.

This paper develops and compares two automated approaches:

- 1) **Traditional image processing** — a hand-crafted pipeline that enhances, binarises, and counts capillary-shaped blobs using filters and morphological tools that do not require any training data.
- 2) **Deep learning (YOLOv8)** — a modern object-detection network that is taught to localise and segment individual capillary loops from annotated example images.

Both approaches are tested on the same dataset of 30 nailfold images covering both normal and pathological cases. By comparing them side by side, the paper aims to answer a practical question: *when does a simple classical method suffice, and when does a neural network add value?*

The rest of the paper is organised as follows. Section II surveys relevant prior work. Section III describes the data. Section IV explains both pipelines in detail. Section V presents results and figures. Section VI interprets the findings, and Section VII concludes.

II. RELATED WORK

A. Enhancing Tubular Structures: CLAHE and Hessian Filters

Capillary loops in nailfold images are thin, elongated, tube-like structures set against a textured skin background. Two classical tools are particularly well-suited to this problem.

CLAHE (Contrast Limited Adaptive Histogram Equalisation) [4] enhances local contrast by equalising the histogram within small image tiles rather than across the whole image. The “clip limit” prevents noise from being amplified. This is important in nailfold images, where contrast varies considerably across the frame depending on skin pigmentation and illumination angle.

Hessian-based vesselness [2] analyses the second-order intensity variations (curvature) at each pixel. At points on a

thin tubular structure, one principal curvature is large (across the vessel wall) while the other is small (along the vessel axis). The Frangi filter formalises this intuition into a *vesselness score* that is high on elongated ridges and low elsewhere. Sato et al. [3] developed a related multi-scale line filter for curvilinear structures in 3-D medical images, further validating the Hessian approach across different imaging modalities.

B. Deep Learning for Capillaroscopy

Convolutional neural networks have been applied to capillaroscopy in several recent works. Avola et al. [5] demonstrated a YOLOv8-based system specifically for nail capillary detection, running on a low-power single-board computer. Their work is the closest precedent to the present study. YOLOv8, developed by Ultralytics [6], is a single-stage detector that outputs bounding boxes, class scores, and instance-level polygon masks in one forward pass, making it well-suited for both counting and visualising individual capillary loops.

C. Gap Addressed by This Paper

Most existing work either applies classical methods *or* deep learning, but rarely compares both on the same data under identical evaluation conditions. Moreover, very few studies report results separately for normal and pathological image classes. This paper addresses both gaps.

III. DATASET

A. Image Classes and Ground Truth

The study uses the **NailFold Capillary Dataset**, which contains 30 widefield capillaroscopy images at a resolution of 1280×1024 pixels. Images are divided into four classes based on clinical pattern:

- **N1** (10 images) — Normal; regular, evenly spaced hairpin loops. Ground-truth counts: 6–9 per image.
- **N2** (3 images) — Normal variant with slightly higher loop density. Ground-truth counts: 9–11.
- **S2** (7 images) — Pathological (active scleroderma pattern); giant capillaries, haemorrhages, reduced density. Ground-truth: 1–5.
- **S3** (10 images) — Late-stage pathological; severe rarefaction, avascular areas, disorganised residual loops. Ground-truth: 2–5.

A key feature of normal images is a clearly visible *black reference line* drawn across the image to mark the measurement zone. Pathological images (S2, S3) do not contain this line, so the full image is used for analysis.

B. Annotation for Deep Learning

For the YOLOv8 pipeline, each capillary loop was manually traced as a polygon using the **Roboflow** [7] annotation tool. Annotations were saved in Pascal VOC XML format and converted to YOLOv8 segmentation format (normalised polygon coordinates in `.txt` files) using a custom script. Two separate models were trained: one for normal images (N1, N2) and one for pathological images (S2, S3). The dataset was split 80:20 into training and validation sets.

Limitation: All annotations were produced by a single non-specialist annotator. Inconsistencies in which structures were labelled as capillaries may affect both model training and the reliability of the ground-truth counts used for evaluation.

IV. METHODOLOGY

Fig. 1 provides a visual overview of the complete system. Both pipelines share the same ROI extraction step (left branch), then diverge into classical processing (left) or deep learning (right).

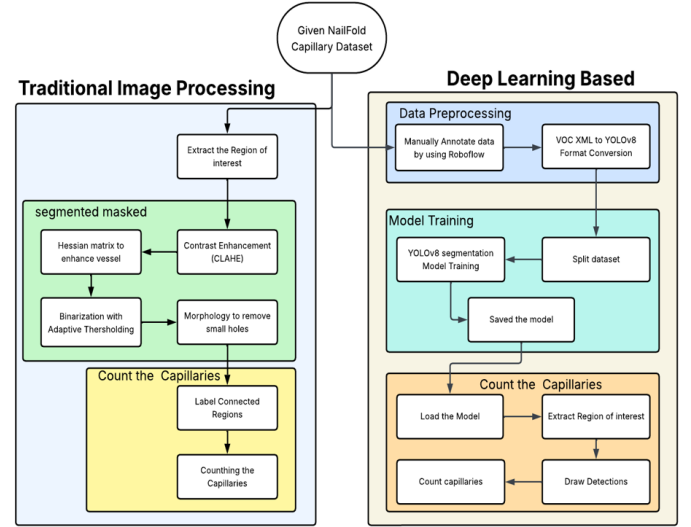


Fig. 1: Overall system architecture. The left branch is the traditional image processing pipeline; the right branch is the YOLOv8 deep learning pipeline. Both share the initial ROI extraction step.

A. Step 0: Region of Interest (ROI) Extraction

Before either pipeline begins, we automatically find and crop the relevant portion of the image — the narrow strip where capillaries are visible above the reference line. This step is illustrated in Fig. 2.

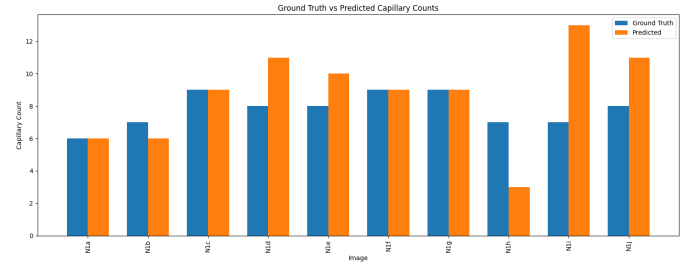


Fig. 2: ROI extraction for a normal (N1) image. Left: the black reference line is detected and highlighted. Centre: the image is rotated to align the line horizontally. Right: the ROI (green rectangle) is defined as a strip above the line (40% of image height).

The procedure works as follows:

- 1) **Detect the black line.** The image is converted to greyscale and an inverse threshold at intensity 30 isolates very dark pixels. External contours are found; the largest one is assumed to be the reference line.
- 2) **Straighten the image.** A straight line is fitted to that contour (`cv2.fitLine`). The angle between the fitted line and horizontal is computed, and the entire image is rotated to remove the tilt (`cv2.warpAffine`).
- 3) **Define the ROI.** The median y -position of the rotated contour gives the line's height. A rectangle spanning 40% of the image height immediately above this line is extracted as the ROI.

For pathological images where no black line exists, the entire image is treated as the ROI.

B. Approach 1: Traditional Image Processing Pipeline

Once the ROI is extracted, it passes through a six-stage classical pipeline. Each stage is illustrated in Fig. 3.

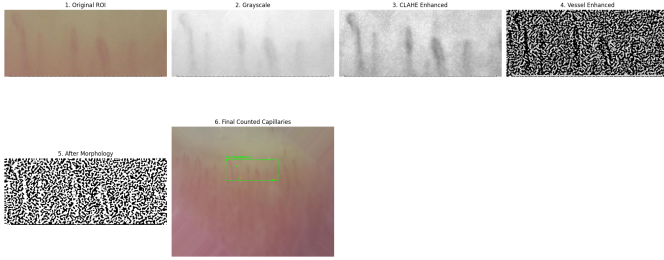


Fig. 3: The six stages of the traditional pipeline applied to image N1a. From left to right: (1) original ROI, (2) green channel extraction, (3) CLAHE contrast enhancement, (4) Hessian vessel enhancement, (5) binary mask after adaptive thresholding, (6) final counting result overlaid on the original.

1) *Stage 1 — Green Channel and CLAHE:* The green channel of the RGB ROI is extracted because haemoglobin absorbs green light more strongly than the surrounding tissue, giving capillaries better contrast in this channel. CLAHE is then applied with a clip limit of 2.5 and a 6×6 tile grid to normalise local contrast while preventing noise amplification [4].

2) *Stage 2 — Hessian Vessel Enhancement:* To bring out the tubular capillary structures, we compute a vesselness score at each pixel using the eigenvalues of the 2×2 Hessian matrix. After Gaussian smoothing with $\sigma = 1.1$ pixels, the second derivatives are estimated with Sobel operators. The two eigenvalues are:

$$\lambda_{1,2} = \frac{(I_{xx} + I_{yy}) \pm \sqrt{(I_{xx} - I_{yy})^2 + 4I_{xy}^2}}{2} \quad (1)$$

Intuitively, on a tubular structure one eigenvalue is large (high curvature across the tube wall) while the other is small (low curvature along the tube axis). The vesselness score from Frangi et al. [2] is:

$$\mathcal{V} = \exp\left(-\frac{R_B^2}{2\beta^2}\right) \left(1 - \exp\left(-\frac{S^2}{2c^2}\right)\right) \quad (2)$$

where $R_B = |\lambda_1/\lambda_2|$ measures how blob-like versus tube-like the structure is, and $S = \sqrt{\lambda_1^2 + \lambda_2^2}$ measures overall structure strength. Parameters $\beta = 0.5$ and $c = 10$ were fixed empirically.

3) *Stage 3 — Adaptive Binarisation:* The vesselness image is binarised using Gaussian adaptive thresholding (block size 15, constant offset 3). Unlike a global threshold, this adapts to local intensity variations across the ROI, producing a cleaner binary mask.

4) *Stage 4 — Morphological Cleaning:* A 2×2 kernel is used for:

- **Opening** (erosion then dilation) — removes isolated noise pixels that do not correspond to real capillaries.
- **Closing** (dilation then erosion) — fills small gaps within capillary structures caused by low-contrast sections [8].

5) *Stage 5 — Connected-Component Counting:* Remaining blobs are labelled using 8-connectivity. Each labelled region is kept only if it satisfies two criteria:

- 1) **Area:** between 32 and 300 pixels (rejects dust specks and merged clusters).
- 2) **Eccentricity** $e > 0.6$: keeps elongated structures; rejects circular artefacts.

Eccentricity is calculated from the fitted ellipse:

$$e = \sqrt{1 - \left(\frac{a_{\min}}{a_{\max}}\right)^2} \quad (3)$$

where $a_{\min}$ and $a_{\max}$ are the minor and major semi-axes. Capillary loops, being narrow and elongated, have e close to 1.

C. Approach 2: YOLOv8 Instance Segmentation

The second approach replaces the hand-crafted steps with a neural network that learns to detect capillaries directly from annotated examples.

1) *Why YOLOv8?:* YOLOv8 [6] is a state-of-the-art single-stage detector that processes an entire image in one forward pass and outputs bounding boxes *plus* polygon masks for each detected instance. The segmentation variant is therefore particularly useful here because we need not only to *count* capillaries but also to *delineate* their exact shapes. We selected `yolov8s-seg.pt` (the “small” variant), which offers a better accuracy–speed trade-off than the nano version for small objects.

2) *Training:* The model was fine-tuned from the pre-trained ImageNet weights. Key hyperparameters are listed in Table I. Because training ran on a CPU-only workstation, the batch size was kept small (4) and early stopping with `patience=30` was used to avoid overfitting on the limited dataset.

3) *Inference and Post-processing:* At test time:

- 1) The trained model is run on the full aligned image.
- 2) Detections whose bounding-box centroids fall *outside* the extracted ROI are discarded.
- 3) Custom Non-Maximum Suppression (NMS) with IoU threshold 0.3 removes duplicate detections of the same capillary.

TABLE I: YOLOv8 Training Hyperparameters

Parameter	Value
Base model	yolov8s-seg.pt
Epochs	100 (early stop at patience 30)
Batch size	4
Input resolution	640 × 640 px
Optimiser	AdamW
Initial LR	0.01
Warmup epochs	3
Rotation augment.	±15°
Scale augment.	±50%
HSV hue shift	0.015
Device	CPU

- 4) The number of remaining detections is the predicted capillary count.

V. RESULTS

A. Evaluation Metrics

We measure accuracy using two complementary metrics:

- **MAE** (Mean Absolute Error): the average absolute difference between predicted and ground-truth counts. Easy to interpret — an MAE of 2.0 means predictions are on average 2 capillaries off.
- **RMSE** (Root Mean Squared Error): penalises large errors more heavily than small ones. A much larger RMSE than MAE indicates a few images with big errors.

B. Traditional Pipeline: N1 Normal Images

Table II shows per-image results for the traditional pipeline on the ten N1 images. The overall MAE is 1.90 and RMSE is 2.74.

TABLE II: Traditional Pipeline — Per-image results on N1

Image	GT	Predicted	Error
N1a	6	6	0
N1b	7	6	-1
N1c	9	9	0
N1d	8	11	+3
N1e	8	10	+2
N1f	9	9	0
N1g	9	9	0
N1h	7	3	-4
N1i	7	13	+6
N1j	8	11	+3
Mean	7.8	8.7	
MAE		1.90	
RMSE		2.74	

Fig. 4 shows a correct result (N1a, error = 0) and a challenging result (N1i, error = +6) side by side, illustrating why some images are harder than others.

C. YOLOv8: All 30 Images

Table III shows per-class MAE and RMSE for the YOLOv8 model. Table IV lists every image individually. The overall MAE is 3.00 and RMSE is 3.91.

Fig. 5 shows predicted versus ground-truth counts for all 30 images as a bar chart, giving an at-a-glance view of where

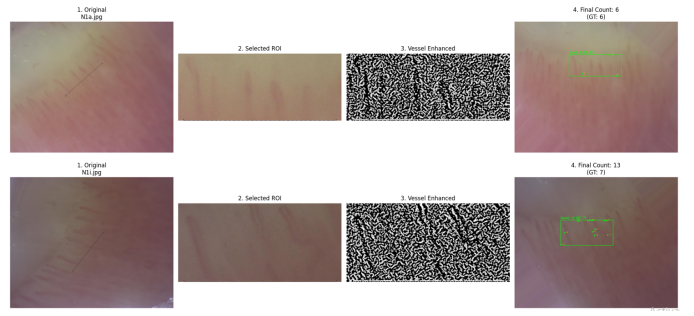


Fig. 4: Traditional pipeline on two images. **Top row (N1a)**: predicted = 6, GT = 6 (error = 0). The pipeline correctly identifies well-separated, high-contrast capillary loops. **Bottom row (N1i)**: predicted = 13, GT = 7 (error = +6). Skin-texture wrinkles are misclassified as capillary structures. Each row shows: original image, selected ROI, vessel-enhanced image, final counted result.

TABLE III: YOLOv8 Per-Class Results

Class	<i>n</i>	Mean GT	MAE	RMSE
N1 (normal)	10	7.8	4.20	4.92
N2 (normal dense)	3	10.0	7.00	7.05
S2 (pathological)	7	2.7	1.14	1.77
S3 (late-stage path.)	10	4.2	1.90	2.12
Overall	30	5.9	3.00	3.91

predictions are close (S2, S3) and where they diverge (N1, N2).

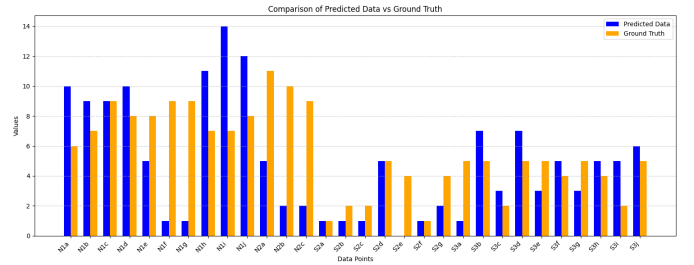


Fig. 5: YOLOv8 predicted counts (blue) vs. ground-truth (orange) for all 30 images. The model tracks ground truth well on pathological classes (S2, S3 — right portion) but shows larger deviations on normal classes (N1, N2 — left portion).

Fig. 6 shows the YOLOv8 output on a normal image (N1c). Red bounding boxes and green polygon masks are drawn around each detected capillary within the cyan ROI rectangle.

Fig. 7 shows the YOLOv8 output on a pathological image (S3j). In this case the capillaries are visually quite different: they are larger, irregularly shaped, and sparse. The model still detects them (GT = 5, predicted = 6) but with lower confidence scores (0.26–0.47) compared to the normal case.

TABLE IV: YOLOv8 Per-Image Results (All 30 Images)

Img	GT	Pr	—E—	Img	GT	Pr	—E—
N1a	6	10	4	S2a	1	1	0
N1b	7	9	2	S2b	2	1	1
N1c	9	9	0	S2c	2	1	1
N1d	8	10	2	S2d	5	5	0
N1e	8	5	3	S2e	4	0	4
N1f	9	1	8	S2f	1	1	0
N1g	9	1	8	S2g	4	2	2
N1h	7	11	4	S3a	5	1	4
N1i	7	14	7	S3b	5	7	2
N1j	8	12	4	S3c	2	3	1
N2a	11	5	6	S3d	5	7	2
N2b	10	2	8	S3e	5	3	2
N2c	9	2	7	S3f	4	5	1
				S3g	5	3	2
				S3h	4	5	1
				S3i	2	5	3
				S3j	5	6	1

GT =

ground truth, Pr = predicted, —E— = absolute error

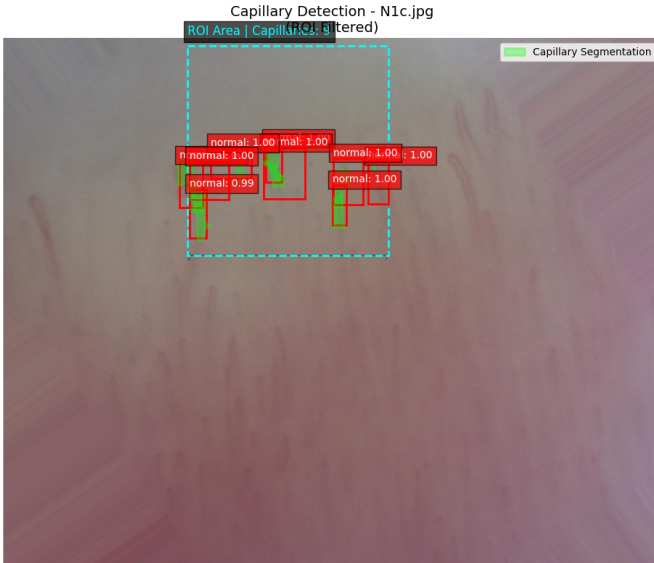


Fig. 6: YOLOv8 detection on normal image N1c (GT=9, predicted=9, error=0). Red boxes mark detected capillaries; green masks show their segmented shapes; the dashed cyan rectangle is the ROI. All nine detections have confidence ≥ 0.99 .

VI. DISCUSSION

A. Why Does Each Method Fail?

Traditional pipeline failures come in two types, both visible in Table II and Fig. 4:

- *Under-counting* (N1h: $pred = 3$, $GT = 7$): Low-contrast capillaries produce a weak Hessian response. After thresholding, these loops fall below the area or eccentricity threshold and are discarded.
- *Over-counting* (N1i: $pred = 13$, $GT = 7$): Skin-texture wrinkles and fold edges are elongated structures that

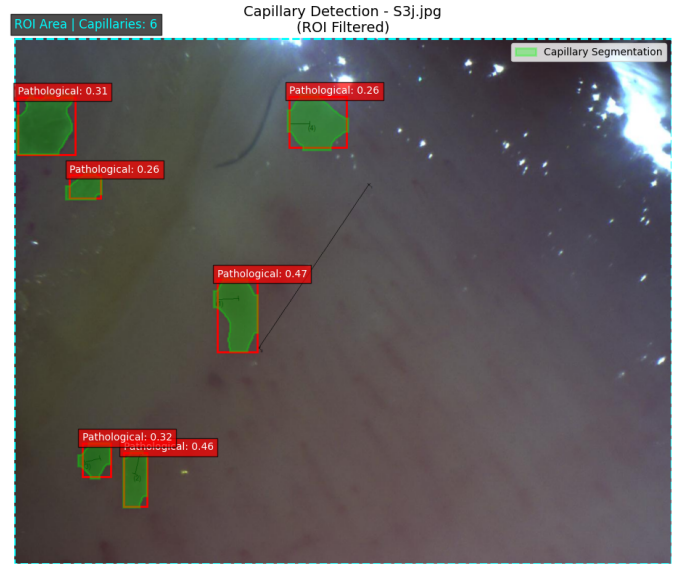


Fig. 7: YOLOv8 detection on pathological image S3j (GT=5, predicted=6, error=1). Lower confidence scores (0.26–0.47) reflect the greater morphological variability of late-stage pathological capillaries compared to the normal case in Fig. 6.

pass the eccentricity filter ($e > 0.6$) and are mistakenly counted as capillaries.

YOLOv8 failures have a different character. On normal images the model over-counts some (N1h: 11, N1i: 14) and under-counts others (N1f: 1, N1g: 1). The zero-or-near-zero predictions for N1f and N1g are particularly suspicious and suggest the model failed to trigger at all on those images, possibly because the contrast or lighting differed from the training set. The N2 class (MAE 7.00) is the worst performer across the whole study: with only 3 training examples and capillary densities nearly double that of N1, the model simply has not seen enough high-density images during training.

On pathological images (S2, S3) the situation reverses: the model achieves MAE values of 1.14 and 1.90, its best results. This is because pathological capillaries are morphologically distinctive (large, blob-like, widely spaced), easy to annotate consistently, and sufficiently different from the skin background to be learned reliably even from a small training set.

B. A Tale of Two Bottlenecks

The traditional pipeline is bottlenecked by **parameter sensitivity**. The eccentricity threshold (0.6), area range (32–300 px), CLAHE clip limit (2.5), and Hessian scale ($\sigma = 1.1$) were all tuned on a single image (N1a) and applied to all others. Because image quality, lighting, and skin tone vary, a fixed parameter set is never ideal for all images.

The YOLOv8 model is bottlenecked by **annotation quality and quantity**. All labels were produced by one non-specialist, which likely introduces inconsistencies in which structures were marked as capillaries. With expert re-annotation and a larger dataset, the model's performance on normal images

would be expected to improve substantially. Evidence for this interpretation comes from the fact that the model performs well on pathological images, where capillary boundaries are more visually unambiguous and thus easier to annotate consistently.

C. Complementarity of the Two Approaches

A practical take-away from the results is that the two methods are *complementary*. The traditional pipeline is reliable and fast on normal images (MAE 1.90 on N1), requires no training data, and produces interpretable intermediate outputs (the enhanced and binary images) that can help a clinician understand why a count was produced. YOLOv8 handles the morphological diversity of pathological images more robustly (S2 MAE 1.14, S3 MAE 1.90) and directly produces instance masks that could support not just counting but also shape-based clinical scoring. A *hybrid* approach — using the traditional method for normal images and YOLOv8 for pathological ones — would combine the strengths of both.

D. Inference Speed

The traditional pipeline runs in under 1 second per image on a standard CPU. YOLOv8 inference takes approximately 535 ms per 512×640 image on CPU (3.3 ms preprocessing, 535.8 ms inference, 546.5 ms postprocessing). GPU acceleration would reduce this to under 20 ms. Both are fast enough for offline clinical use; real-time use would require a GPU.

E. Limitations

- 1) **Small dataset.** 30 images spread across 4 classes means each class is evaluated on 3–10 examples, making statistical conclusions fragile.
- 2) **Non-specialist annotations.** Ground-truth labels and training polygons were produced by a student, not a rheumatologist.
- 3) **No cross-validation.** Results are from a single train/val split.
- 4) **Traditional method tested only on N1.** A direct comparison on all 30 images is not available for the classical pipeline.
- 5) **No mAP metrics.** YOLOv8 segmentation mAP metrics were not recorded during training; only count-based MAE and RMSE are reported.

F. Future Work

Priority recommendations for future work are:

- **Expert re-annotation** by a certified capillaroscopy specialist.
- **Dataset expansion:** at least 200 images per class for robust deep learning training.
- **Multi-scale Hessian filtering:** spanning $\sigma \in \{0.5, \dots, 3\}$ for better robustness across varying capillary widths.
- **Hybrid pipeline:** use the traditional method as an attention mask to guide YOLOv8 inference in challenging low-contrast regions.
- **Newer architectures:** explore SAM (Segment Anything Model) or MedSAM adapted for capillaroscopy.

VII. CONCLUSION

This paper compared two automated approaches for capillary counting in nailfold capillaroscopy images. The key findings are:

- The **traditional pipeline** (CLAHE + Hessian vesselness + adaptive thresholding + connected components) is simple, fast, interpretable, and achieves a competitive MAE of **1.90** on normal images without any training data. Its main weakness is sensitivity to contrast variations and non-capillary elongated structures.
- **YOLOv8 segmentation** excels on pathological images (S2 MAE = **1.14**, S3 MAE = **1.90**) where morphologically distinct capillaries provide a strong training signal, but struggles on high-density normal images (N2 MAE = 7.00), reflecting the limited and non-expert training annotations.
- Both approaches are primarily constrained by **data quality**, not by algorithmic limitations. Expert annotation and a larger dataset would be expected to significantly close the gap between current performance and clinical requirements.

Together, the two methods form a complementary toolkit: the traditional approach is well-suited to normal-capillary screening, while YOLOv8 offers greater morphological flexibility for pathological assessment. A hybrid system combining both, trained on expert-annotated data, represents a promising direction toward clinically deployable automated capillaroscopy analysis.

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